TauTimp — Teoria Onduscularǎ: Quantization of Space, the Primordial Black Hole, Muon Neutrino Flux, and Wheeler's Delayed-Choice Experiment

1 TauTimp — Teoria Onduscularǎ: Quantization of Space, the Primordial Black Hole, Muon Neutrino Flux, and Wheeler's Delayed-Choice Experiment

Author: Viorel-Mihai Popescu **Framework:** Teoria Onduscularǎ (TauTimp — Temporal Undulatory Theory)
Version: v2.0 — 21 August 2026 (unifies the PBH/quantic-space note, the muon-neutrino-flux derivation, and the Wheeler/TIQM note into a single document) **Compatible with:** TauTimp Unified Framework v1.0; TauTimp Cosmology of the Local Universe v1.0

1.1 Abstract

Within the TauTimp (Temporal Undulatory Theory) framework, space is intrinsically quantized from the elementary length scale of the quark ($\ell_q \approx 0.43 \times 10^{-18}$ m) up to the boundary of the Local Universe ($2R_s \approx 17.72738$ Gly), and every observable sector — gravity, space, time, charge, heat, mass — obeys one variable-coefficient spherical wave equation of the form $(R_X/r)\Delta\psi = c^{-2}\,\partial_t^2\psi$, distinguished only by its boundary radius R_X. This document unifies four previously separate threads of that framework into a single, non-duplicated derivation:

1. The **quantization of space** and the **onduscular wave equation** for the muon-neutrino-mediated space field $\psi(r,t)$, solved exactly via separation of variables and reduction to the Airy equation, giving a discrete eigenvalue spectrum λ_n, ω_n and the geometric identity $2R_s = R \cdot d_N/d_q$ linking the Local Universe radius to the nucleon-to-quark size ratio.
2. The **muon neutrino flux** $P(r,t)$ and $F(r)$, obtained as the d'Alembert solution of the companion 1-D wave equation, whose normalization over the onduscular sphere yields an **analytic derivation of** H_0 from R_s and $r_{\min}$ alone — a postdiction matching the Planck 2018 value $H_0 = 67.41$ km/s/Mpc.
3. The **Primordial Black Hole (PBH)** — the singularity-free initial state of the Local Universe — with its Schwarzschild radius, its mass estimate via a Sagittarius A* volumetric analog, its internal Temporal Event Horizon and inflation-trigger criterion, its resolution of the central-singularity problem through space granulation, and the resulting matter–antimatter asymmetry.
4. **Wheeler's delayed-choice gedanken experiment** — realized with a single ultra-cold helium atom by Truscott and collaborators — reinterpreted through Cramer's Transactional Interpretation of Quantum Mechanics (TIQM), with the TauTimp-specific assignment of two distinct propagation speeds: the observable **offer wave** at c, and the unobservable **confirmation wave** (the longitudinal entanglement particle) at $V = c \times 10^9$.

The unifying thread across all four parts is the single onduscular field equation and its two companion carriers: the muon neutrino ν_μ, which materializes space and whose flux fixes H_0 and seeds the PBH's own explosive origin, and the wave–particle "instantaneous quantum jump" (§ Part III), which is the same duality that Wheeler's experiment probes directly in the laboratory.

1.2 Table of Contents

Part I — Quantization of Space and the Onduscular Wave Equation 1. The Quantic Nature of Space 2. The Onduscular Wave Equation of Quantic Space 3. Exact Solution: Separation of Variables and the Airy-Equation Quantization 4. The General Field $H(r,\theta,\phi,t)$ and the Geometric Identity $2R_s = R \cdot d_N/d_q$ 5. Equivalence Between Gravity and Quantic-Space Equations

Part II — Muon Neutrino Flux and the Derivation of H_0 6. The Flux Wave Equation for $P(r,t)$ 7. Muon Neutrino Flux $F(r)$ and the Analytic Derivation of H_0 8. Space Expansion from the Integral Over Cosmic Scales 9. Summary of Key Field-Theory Results

Part III — Wave–Particle Duality and the Quantum Jump 10. Time-Quantized Microparticle and the Instantaneous Quantum Jump 11. Double-Slit Interpretation

Part IV — The Primordial Black Hole: Initial State of the Local Universe 12. Initial Configuration and the Hartle–Hawking No-Boundary Analog 13. PBH Schwarzschild Radius 14. The Sagittarius A* Analog and Mass Estimate 15. Temporal Event Horizon and the Inflation Criterion 16. Space Granulation and the Absence of Singularity 17. Matter–Antimatter Asymmetry from Quark Spin Distribution 18. Observational Anchor: GN-z11 and the JWST Boundary

Part V — Wheeler's Delayed-Choice Experiment and the Transactional Interpretation of Quantum Mechanics 19. Wave–Particle Duality and the Role of Measurement 20. Temporal Ionisation and Neutrino Energy Thresholds 21. Wheeler's Delayed-Choice Experiment 22. The Transactional Interpretation of Quantum Mechanics (TIQM) 23. Mathematical Model 24. Physical Consequences 25. Key Insight 26. Summary of Numerical Values

Part VI — Parameter Tables 27. Fundamental TauTimp Cosmological and Field Parameters 28. Primordial Black Hole Parameters 29. Galactic Black Hole (Sagittarius A*) Parameters 30. Quantum Limits and Wheeler/TIQM Values

References

2 Part I — Quantization of Space and the Onduscular Wave Equation

2.1 1. The Quantic Nature of Space

Within the TauTimp framework, **space is not a continuous manifold** but a quantized medium spanning a finite range of scales:

- **Lower bound (microscopic):** $\ell_q \approx 0.43 \times 10^{-18}$ m — the quark linear dimension (cf. arXiv:1604.01280).
- **Upper bound (cosmological):** $R_{LU} = 2R_s \approx 1.6771 \times 1026$ m ≈ 17.72738 Gly — the boundary of the Local Universe.

The Local Universe is defined as the maximal finite spherical domain from which no information — photonic, gravitonic, or onduscular — can escape:

$$R_{LU} = 2R_s, \qquad \text{nothing escapes: } \gamma, \nu, \text{photons reflected at } r = R_{LU} \qquad (1)$$

The Local Universe is itself embedded within the **Big Universe**, estimated at ~ 37 Tly $\approx 400 \times R_{LU}$, the latter playing no direct role in dynamics within R_{LU} due to the impenetrability of the muon onduscular horizon.

Generative pair: Higgs boson and muon neutrino. The materialization of space is driven by the bonded pair $\{H^0, \nu_\mu\}$: **mass** generates **gravity**, mediated by the Higgs boson H^0; **gravity** bends **space**, materialized through the muon neutrino ν_μ. The muon neutrino's onduscular field $\psi(r,t)$ is the dynamical variable of the quantic-space equation derived below.

2.2 2. The Onduscular Wave Equation of Quantic Space

The fundamental TauTimp equation governing the quantic-space field is the variable-coefficient spherical wave equation: $\boxed{\frac{R_s}{r}\,\Delta\psi(r,t) = \frac{1}{c^2}\frac{\partial^2\psi(r,t)}{\partial t^2}}$ (2)

where $R_s = 8.3855 \times 1025$ m is the **muon onduscular radius** (different from the graviton-sector parameter $R = 4.2355 \times 1022$ m), and Δ is the Laplacian in spherical coordinates.

For the closely related **gravity-sector** equation (graviton, tau neutrino, magnetron, electron) the parameter R replaces R_s: $\frac{R}{r}\,\Delta\psi(r,t) = \frac{1}{c^2}\frac{\partial^2\psi(r,t)}{\partial t^2}$ (3)

Equation (3) coincides with the canonical TauTimp Unified Framework equation. The two equations have **identical functional form** but distinct boundary parameters, encoding the disparity between the gravitational reach ($\sim 4.48\ Mly$) and the spatial extent of the Local Universe ($\sim 17.73\ Gly$).

Effective propagation speed. Equation (2) describes wave propagation with a space-dependent effective velocity:

$$v_{\text{eff}}(r) = c\sqrt{\frac{R_s}{r}} \qquad (4)$$

This diverges as $r \to 0$ and vanishes as $r \to R_{LU}$, encoding the onduscular confinement and the regularization of the centre.

Comparison with parent equations:

Equation	Form
Schrödinger	$\frac{2m}{\hbar^2}[E - U(r)]\psi + \Delta\psi = \frac{i}{\hbar}\partial_t\psi$
Klein–Gordon	$\frac{2m}{\hbar^2}\psi + \Delta\psi = \frac{1}{c^2}\partial_t^2\psi$
TauTimp (graviton)	$(R/r)\Delta\psi = c^{-2}\,\partial_t^2\psi$
TauTimp (muon, quantic space)	$(R_s/r)\Delta\psi = c^{-2}\,\partial_t^2\psi$

For atomic radii $r \sim r_{\text{Bohr}}$, the TauTimp equation reduces (in the appropriate limit) to the Schrödinger equation; at galactic and cosmological distances, it correctly predicts the wave function evolution that classical quantum mechanics cannot.

2.3 3. Exact Solution: Separation of Variables and the Airy-Equation Quantization

For the **spherically symmetric** case ($\psi = f(r,t)$, no θ, ϕ dependence), the spherical Laplacian reduces to $\Delta f = f_{rr} + \frac{2}{r}f_r$, and equation (2) becomes: $\boxed{\frac{R_s}{r}\left(f_{rr} + \frac{2}{r}f_r\right) = \frac{1}{c^2}f_{tt}}$ (5)

Separation of variables. Writing $f(r,t) = g(r)\,u(t)$ and dividing by $g\,u$ gives two ordinary differential equations, with separation constant $-\lambda^2$: $u'' + \lambda^2 c^2 u = 0$ (T) ; $g'' + \frac{2}{r}g' + \frac{\lambda^2 r}{R_s}g = 0$ (R)

Time equation — exact solution. Equation (T) is simple harmonic:

$$\boxed{u(t) = A\cos(\lambda c\, t) + B\sin(\lambda c\, t)} \qquad (6)$$

with angular frequency $\omega = \lambda c$.

Radial equation — reduction to the Airy equation. Substituting $g(r) = r^{-1/2}\phi(r)$ (motivated by the Bessel-type structure) and multiplying by $r^{1/2}$, equation (R) becomes:

$$\phi'' + \left(\frac{\lambda^2 r}{R_s} - \frac{1}{4r^2}\right)\phi = 0$$

For $r \gg \lambda^{-1}$ (wave-dominated regime), the $1/(4r^2)$ term is negligible. Defining $\xi = \alpha r$, $\alpha = (\lambda^2/R_s)^{1/3}$, the equation reduces to the **Airy equation**: $\boxed{\frac{d^2\phi}{d\xi^2} = \xi\,\phi}$ (7)

with general solution $\phi(\xi) = C_1\mathrm{Ai}(\xi) + C_2\mathrm{Bi}(\xi)$.

Boundary conditions and quantization. The boundary condition $f(2R_s, 0) = 0$ requires $\phi(\xi_n) = 0$ at $\xi_n = \alpha \cdot 2R_s$. Physical (decaying at infinity) solutions require $C_2 = 0$ (since Bi grows exponentially), so $\phi(\xi) = C_1\mathrm{Ai}(\xi)$, and the boundary condition selects the n-th zero a_n of Ai:

$$\alpha_n \cdot 2R_s = -a_n \quad (a_n < 0,\ n = 1,2,3,\dots)$$

$$\boxed{\lambda_n = \frac{1}{2R_s}(-a_n)^{3/2}\sqrt{R_s} = \frac{(-a_n)^{3/2}}{2\sqrt{R_s}}} \qquad (8)$$

The first few Airy zeros: $a_1 \approx -2.3381$, $a_2 \approx -4.0879$, $a_3 \approx -5.5206$.

Radial eigensolutions: $g_n(r) = r^{-1/2}\,\mathrm{Ai}(\alpha_n r)$, $\alpha_n = \left(\frac{\lambda_n^2}{R_s}\right)^{1/3}$ (9)

Full scalar field: $\boxed{f(r,t) = \sum_{n=1}^{\infty} r^{-1/2}\,\mathrm{Ai}(\alpha_n r)[A_n\cos(\omega_n t) + B_n\sin(\omega_n t)], \quad \omega_n = \lambda_n c}$ (10)

Coefficients A_n, B_n are fixed by the initial conditions $f(r,0)$ and $f_t(r,0)$.

2.4 4. The General Field $H(r,\theta,\phi,t)$ and the Geometric Identity $2R_s = R \cdot d_N/d_q$

For the **general non-symmetric** case, the field expands in real spherical harmonics:

$$H(r,\theta,\phi,t) = \sum_{\ell=0}^{\infty}\sum_{m=-\ell}^{\ell} f_{\ell m}(r,t)\,Y_\ell^m(\theta,\phi) \qquad (11)$$

where each radial mode satisfies the ℓ-th radial equation, now including the centrifugal barrier:

$$g_\ell'' + \frac{2}{r}g_\ell' + \left(\frac{\lambda^2 r}{R_s} - \frac{\ell(\ell+1)}{r^2}\right)g_\ell = 0 \qquad (12)$$

For $\ell = 0$ (spherical symmetry), $Y_0^0 = (4\pi)^{-1/2}$ and Part I §3 is recovered exactly.

The three boundary conditions. The muon onduscular wave function $\psi(r,t)$ is uniquely determined by:

(A) $\psi(2R_s, t) = 0$ for all t — equivalently $g(2R_s) = 0$ after separation — encoding the impenetrability of the Local Universe boundary.

(B) $g'(0) = R_s$ — the initial radial derivative at the origin.

(C) Speed of interaction equals c, expressed via the divergence operator $\mathrm{div}[\psi(r,ct)] = 1$, i.e. $\partial_t[u(t)g(r)] + \partial_r[u(t)g(r)]\,\frac{dr}{dt} = c$.

These three conditions uniquely select the basis of separated solutions (10). Condition (A) is the same quantization condition applied in §3; conditions (B)–(C) fix the origin behaviour and the interaction speed. Separately, the muon-flux normalization $\int_{r_{\min}}^{2R_s} F(r)\,dr = c/H_0$ derived in Part II §7 provides the complementary condition that fixes the overall amplitude of the flux sector — the two boundary-value problems (field ψ and flux P) are companions sharing the same domain $[r_{\min}, 2R_s]$ but are solved independently.

The geometric identity. A central TauTimp geometric identity relates the Local Universe radius to the nucleon-to-quark size ratio: $\boxed{2R_s = R \cdot \frac{d_N}{d_q}}$ (13)

Numerical substitution: $2R_s = 2 \times 4.2355 \times 1022 \times \frac{0.841419}{0.43\times10^{-3}} = 16.7710 \times 1025$ m $=$ 17.72738 Gly (14)

where $d_N = 0.841419$ fm is the nucleon (proton) charge radius and $d_q = 0.43$ am $= 0.43 \times 10^{-18}$ m is the quark linear dimension. Equation (14) provides the **first-principles cosmological prediction** for the size of the Local Universe — a number derived from sub-femtometre physics yet measured in giga-light-years.

The same equation, with parameter R, describes the **graviton, the tau neutrino, the magnetron, and the electron**. Only the **muon neutrino** (and hence the quantic-space field) has the larger boundary R_s.

2.5 5. Equivalence Between Gravity and Quantic-Space Equations

The scaling ratio. The dimensionless ratio between the two TauTimp boundary parameters is:

$$\frac{R_s}{R} = \frac{8.3855 \times 1025}{4.2355 \times 1022} \approx 1979.8 \qquad (15)$$

Coordinate rescaling transformation. Under the substitution $r \to r/1979.8$, equation (3) (graviton sector) is mapped onto equation (2) (muon/space sector). Thus:

- The effective wave functions are **proportional**, not equal.
- The **curvature of space** is approximately 1980 times **smaller** than the curvature of the gravitational field at any given physical radius.

- Space and gravity are **manifestations of the same wave equation**, but they have different couplings and different carriers: ν_τ (graviton) and ν_μ (space-mediator) — distinct in structure and coverage.

This is the mathematical content of the often-cited observation that "in general relativity, the gravity field and space are the same thing to Einstein, but they are not": the two equations are **isomorphic** but **not identical**, and only after the scaling (15) do they coincide.

Linear superposition of onduscular coordinates. The full set of TauTimp coordinates with spherical symmetry — Gravity (ν_τ), Phonon, 3D Space (ν_μ), Time (τ), Electric Charge (electron), Heat (Thermal particle), Mass coordinate (Bottom quark and Higgs boson) — obey **linearly ordered** wave equations of the universal form $(R_X/r)\Delta\psi = c^{-2}\,\partial_t^2\psi$ with different R_X. **Superposition is therefore valid** across the seven sectors. The Riemannian generalization (general relativity) applies to each spherically symmetric coordinate: each bends the "space–time" of the multidimensional Riemann manifold.

3 Part II — Muon Neutrino Flux and the Derivation of H_0

3.1 6. The Flux Wave Equation for $P(r,t)$

The differential muon neutrino flux $P(r,t)$ satisfies its own 1-D wave equation in the radial variable:

$$\frac{1}{c^2}P_{tt} - P_{rr} = 0 \qquad \text{(W)}$$

with **boundary condition** $P(2R_s,0) = 0$ and **normalization**: $\int_{r_{\min}}^{2R_s} P\,(r,0)\,dr = 1$ (16)

Degree-2 polynomial solution at $t = 0$. For a quadratic $P_0(r) = p_2r^2 + p_1r + p_0$, $P_{rr} = 2p_2 = \text{const}$, so equation (W) is satisfied at $t = 0$ for any such polynomial. The physically natural, bell-shaped **factored form** — vanishing at the inner boundary $r_{\min}$ and the outer boundary $2R_s$, positive in between — is:

$$\boxed{P(r,0) = C\,(r - r_{\min})(2R_s - r)} \qquad (17)$$

General time evolution. Since (W) is the ordinary 1-D wave equation, the general solution via d'Alembert, for initial velocity $P_1 = \partial_t P|_{t=0} = 0$, is:

$$P(r,t) = 1/2\,[P_0(r+ct) + P_0(r-ct)]$$

$$\boxed{P(r,t) = C[(r - r_{\min})(2R_s - r) - c^2t^2]} \qquad (18)$$

3.2 7. Muon Neutrino Flux $F(r)$ and the Analytic Derivation of H_0

Definition. The differential surface flux at $t = 0$:

$$\boxed{F(r) = \frac{P(r,0)}{4\pi r^2} = \frac{C}{4\pi}\cdot\frac{(r - r_{\min})(2R_s - r)}{r^2}} \qquad (19)$$

Physical justification. The flux $F(r)$ integrated over a radial shell from $r_{\min}$ to the Hubble radius $R_H = c/H_0$ gives:

$$\int_{r_{\min}}^{2R_s} F\,(r)\,dr = \frac{c}{H_0} \qquad (20)$$

The flux-integrated column depth equals the Hubble horizon scale $R_H = c/H_0$. In the onduscular framework, the boundary $r = 2R_s$ marks the edge of the onduscular sphere; the Hubble law $v = H_0r$ at $r = c/H_0$ marks where the recession velocity equals c. Equation (20) connects the local neutrino flux distribution to the cosmic expansion rate — a core prediction of TauTimp, consistent with the observation $2R_s \approx 17.73$ Gly $\approx 1.222\,R_H$, placing the onduscular boundary just beyond the Hubble horizon.

Analytic evaluation of the integral.

$$\int_{r_{\min}}^{2R_s} \frac{(r - r_{\min})(2R_s - r)}{r^2}\,dr = \left[(r_{\min} + 2R_s)\ln\frac{2R_s}{r_{\min}} - 2(2R_s - r_{\min})\right] \equiv \mathcal{G}$$

Numerically: $\mathcal{G} = 3.7113 \times 1026$ m.

Determination of coefficient C. From $\frac{C}{4\pi}\cdot\mathcal{G} = \frac{c}{H_0}$: $\boxed{C = \frac{4\pi c}{H_0\,\mathcal{G}}}$ (21)

With $H_0 = 67.41$ km/s/Mpc (Planck 2018): $C = 4.6465$ m^{-2}.

Expanded polynomial coefficients of $P(r,0) = -Cr^2 + C(r_{\min} + 2R_s)r - C\,r_{\min}\cdot 2R_s$:

Coefficient	Expression	Value (SI)	
p_2	$-C$	-4.6465 m^{-2}	
p_1	$C(r_{\min} + 2R_s)$	7.9202×1026	m^{-1}
p_0	$-C\,r_{\min}\cdot 2R_s$	-2.1381×1051	

In Gly units ($\tilde{r} = r/\text{Gly}$, $\tilde{C} = C\cdot\text{Gly}^2$): $\tilde{p}_2 = -4.159 \times 1050$, $\tilde{p}_1 = 7.493 \times 1051$, $\tilde{p}_0 = -2.138 \times 1051$.

Hubble constant from the integral — analytic formula. Setting $\int_{r_{\min}}^{2R_s} F\,(r)\,dr = c/H_0$ and solving for H_0:

$$\boxed{H_0 = \frac{4\pi c}{C\left[(r_{\min} + 2R_s)\ln\frac{2R_s}{r_{\min}} - 2(2R_s - r_{\min})\right]}} \qquad (22)$$

With $r_{\min} = 0.290$ Gly and $R_s = 8.8632$ Gly: $H_0 = 67.41$ km/s/Mpc, in excellent agreement with the Planck 2018 result ($H_0 = 67.4 \pm 0.5$ km/s/Mpc). This is a **postdiction** of TauTimp: the onduscular parameters R_s and $r_{\min}$ fix the Hubble constant.

3.3 8. Space Expansion from the Integral Over Cosmic Scales

For the flux integrated from $r_{\min}$ to an upper limit $R_{\max}$:

$$\Phi(R_{\max}) \equiv \int_{r_{\min}}^{R_{\max}} F\,(r)\,dr$$

At $R_{\max} = 2R_s \approx 17.73$ Gly $\approx 1.222\ R_H$: $\Phi(2R_s) = c/H_0 = R_H$. This means: **the spatial integral of the onduscular flux over the full domain equals the Hubble radius** — the flux profile, when integrated, samples the same length scale as the expansion horizon. The Hubble constant thus emerges from purely geometric onduscular parameters, $H_0 = 4\pi c/\mathcal{G}(R_s, r_{\min})$, constituting a **derivation** of H_0 from R_s and $r_{\min}$ that connects cosmological expansion to the onduscular sphere radius.

3.4 9. Summary of Key Field-Theory Results

Result	Formula	Value
Factored $P(r,0)$	$C(r - r_{\min})(2R_s - r)$	—
Analytic $\mathcal{G}$	$(r_{\min} + 2R_s)\ln(2R_s/r_{\min}) - 2(2R_s - r_{\min})$	3.7113×1026 m
C	$4\pi c/(H_0\mathcal{G})$	4.6465 m^{-2}
$\int F\,dr$	c/H_0	1.3723×10^{26} m
H_0 (derived)	$4\pi c/(C\mathcal{G})$	67.41 km/s/Mpc
λ_n (quantization)	$(-a_n)^{3/2}/(2\sqrt{R_s})$	discrete spectrum
ω_n	$\lambda_n c$	discrete frequencies

4 Part III — Wave–Particle Duality and the Quantum Jump

4.1 10. Time-Quantized Microparticle and the Instantaneous Quantum Jump

The TauTimp interpretation of wave–particle duality is **temporal**:

> A particle in motion is, for one time quantum, a microparticle (massive corpuscle); for the next time quantum, a wave of probability. Between successive time quanta it executes an **instantaneous quantum jump**.

The mediator of the mass redistribution at each particle-to-wave transition is the muon neutrino ν_μ — the same carrier of the quantic-space field derived in Part I — emitted at the instant of the transition and carrying the distribution of mass to the next position.

4.2 11. Double-Slit Interpretation

In the double-slit experiment, the electron interferes with itself **in the time quanta during which it is a wave**. If the slit width is of the order of the de Broglie wavelength of the incident particle, the wave-quantum portion of the trajectory undergoes interference; if not, only the particle-quantum portion is recorded.

The spatial quantum is the quark dimension, $\ell_q \approx 0.43 \times 10^{-18}$ m. The temporal quantum (for photons, mediated by ν_τ) is $T_q \approx 1.765 \times 10^{-19}$ s — the time required for a photon to traverse the Bohr radius $a_0 = 0.529$ Å at speed c.

The Heisenberg uncertainty principle currently limits the experimental resolution to $\sim 10^{-18}$ s, so direct confirmation of the temporal quantum lies just below current measurement capability. This same duality is put to direct experimental test in Part V, via Wheeler's delayed-choice experiment.

5 Part IV — The Primordial Black Hole: Initial State of the Local Universe

5.1 12. Initial Configuration and the Hartle–Hawking No-Boundary Analog

The Local Universe began as a **Primordial Black Hole (PBH)** of approximately one mile in characteristic linear dimension ($\sim$ 1.6 km), consisting of a dense **quark–gluon plasma** under extreme gravitational confinement. The PBH is a **singularity-free initial state**: the minimum inter-quark distance imposed by space granulation (§16) prevents infinite density.

In the spirit of the Hartle–Hawking no-boundary proposal, the Local Universe is **finite in size** and **without spatial boundary in the topological sense** — the boundary condition $\psi(2R_s) = 0$ is dynamical, not geometric. The universe began from the PBH and underwent inflationary expansion mediated by the appearance of τ and μ neutrinos at $t_\star \approx$ 290 Myr, generating Time and Space as observable quantities.

5.2 13. PBH Schwarzschild Radius

Applying the Schwarzschild formula to the PBH: $R_{\mathrm{Sch}}^{(\mathrm{PBH})} = \frac{2GM_{\mathrm{PBH}}}{c^2} \approx R = 4.2355 \times 1022$ m $\approx$ 4.48 Mly (23)

This **coincides** with the TauTimp fundamental radius R, which is therefore identified with the PBH Schwarzschild radius and represents the **inflationary limit** in both space and time. [updated 7 Nov 2026: with R = 4.2355·10²² m, 2GM_PBH/c² for M_PBH ≈ 1.31·10¹⁹ M⊙ gives ≈3.87·10²² m ≈ 0.91 R (it was ≈0.99 R for R = 3.9228·10²² m)]

A consequence: stars near the PBH are attracted and engulfed, but **the PBH itself is not gravitationally pulled by external stars** — the graviton cannot escape from the PBH (escape velocity $> c$). Hence:

- The supermassive black hole at the centre of the Milky Way may **spin** but does not **translate** under gravitational attraction from nearby stars.
- A **pair of black holes** do not attract each other directly; what attracts is the **accretion disk** of each. This produces the long, slow "gravitational negotiation" observed astronomically.

5.3 14. The Sagittarius A* Analog and Mass Estimate

Sagittarius A* parameters. The supermassive black hole at the centre of the Milky Way, **Sagittarius A***, has mass $M_{\text{Sgr A*}} \approx 4 \times 10^6\, M_\odot$ and Schwarzschild radius: $r_s^{\text{Sgr A*}} = \frac{2GM_{\text{Sgr A*}}}{c^2} = 1.217 \times 10^{10}$ m (24)

The quark soup at the centre has a characteristic size between a basketball and a soccer ball — diameter ~ 22 cm, radius ~ 11.2 cm.

Volumetric scaling to the PBH. The volumetric scaling factor between the Sagittarius A* quark soup ($r \sim 11.2$ cm) and the PBH ($r \sim 1$ mile ≈ 1663 m):

$$\zeta = \left(\frac{1663}{0.112}\right)^3 \approx (1.4848 \times 10^4)^3 \approx 3.27357 \times 10^{12} \qquad (25)$$

PBH mass estimate. Assuming the same quark density in both objects:

$$\boxed{M_{\text{PBH}} \approx \zeta \times M_{\text{Sgr A*}} = 3.27357 \times 10^{12} \times 4 \times 10^6\, M_\odot \approx 1.31 \times 10^{19}\, M_\odot} \qquad (26)$$

This is the TauTimp prediction for the **initial mass of the Local Universe** prior to its onduscular expansion.

5.4 15. Temporal Event Horizon and the Inflation Criterion

Internal horizon structure. Inside the Gravitational Event Horizon $r_s^{\text{Sgr A*}}$ of the Milky Way's black hole, TauTimp predicts a **Temporal Event Horizon (Te.Ev.Ho.)** with radius:

$$\boxed{R_{\tau\text{EvHo}} = 75.587 \text{ m}} \qquad (27)$$

Within this inner sphere, **time runs backwards**: the tau neutrino ν_τ has spin projection $-1/2$ (instead of the standard $+1/2$), and "younger" quarks emerge from the quark soup.

Inflation criterion. When the quark soup compresses such that its characteristic radius reaches:

$$\boxed{R_{s\text{EvHo}} = 22\, R_{\tau\text{EvHo}} \approx 1663 \text{ m}} \qquad (28)$$

the space within this region **collapses inward**, accumulates critical energy density, and **explodes outward** in a Big-Bang-like event. This is the **TauTimp inflationary trigger**: any sufficiently large black hole evolves through this radius and seeds a new Local Universe. The fact that $R_{s\text{EvHo}} \approx 1663$ m ~ 1 mile provides the physical justification for the PBH characteristic dimension of §12 — the PBH is precisely a quark soup at the inflationary trigger radius.

5.5 16. Space Granulation and the Absence of Singularity

Minimum inter-quark distance. Within the centre of the Milky Way's black hole, the minimum inter-quark distance imposed by the granulation of space is:

$$\ell_{\text{min}}^{\text{Sgr A*}} = 3.524 \times 10^{-25} \text{ m} \qquad (29)$$

corresponding to a per-quark mass of 2.5559×10^{-31} kg packed into a uniform-density core. The light-traversal time across this distance is: $\tau_{\text{cross}} = \frac{\ell_{\text{min}}^{\text{Sgr A*}}}{c} = 1.17548 \times 10^{-33}$ s (30)

Gravity is large but finite. Equations (29)–(30) imply that the gravitational force at the centre of any black hole is **very large but finite** — there is **no singularity** at the geometric origin. The TauTimp resolution of the central singularity is therefore **kinematic** (granulation), not topological.

Photonic origin velocity. A photon emitted at the centre would traverse the inter-quark distance in ~ 1.18×10^{-33} s, well below the Planck time ($t_P \approx 5.4 \times 10^{-44}$ s); however, the photon is **trapped** inside the gravitational horizon (escape velocity $> c$) and only propagates outward upon emission of an accretion-disk photon at a much larger radius.

5.6 17. Matter–Antimatter Asymmetry from Quark Spin Distribution

Spin projection distribution at the Big Bang. The asymmetry between matter and antimatter in the Local Universe is **not** generated by CP-violating processes alone but emerges from the **statistical distribution of quark spin projections** at the moment of the PBH explosion:

- The **majority** of quarks emerge with spin projection $+1/2$ along the positive time arrow → ordinary matter.
- A **minority** emerge with spin projection $-1/2$ along the negative time arrow → antimatter.

Quark mass asymmetry. In parallel, the quark mass asymmetry under spin reversal — the bottom quark b (~ 4.18 GeV) being heavier than $\bar{b}$ in the relevant decay channels — provides a multiplicative factor that further amplifies matter dominance.

Quantitative implication. The combination of quark spin distribution and mass asymmetry, both rooted in the PBH initial conditions, accounts for the present-day matter-to-antimatter ratio of ~ 10^{10}:1 inferred from CMB and baryogenesis observations. A detailed numerical treatment requires solving the onduscular wave equation (2) with the appropriate boundary conditions at $t = 0^+$ post-PBH explosion.

5.7 18. Observational Anchor: GN-z11 and the JWST Boundary

Cosmological timescale. We adopt the standard age of the Local Universe $t_U = 13.79$ Gyr and the expansion driven by muon neutrino propagation at the speed of light. The photon event horizon is therefore $2R_s = 17.72738$ Gly in

comoving terms, while the light-travel distance including expansion is ~ 92–94 Gly. The first stars appeared at $t_\star \approx$ 290 Myr, coinciding with the emergence of τ and μ neutrinos (and hence time and space as observable quantities).

Galaxy GN-z11 parameters. Using the Wright cosmology calculator with $\Omega_M = 0.018$, $\Omega_\Lambda = 1$, $H_0 = 69.6$ km/s/Mpc, the parameters of GN-z11 ($z = 11.1$, Oesch et al. 2016) at the time of emission are:

Quantity	Value
Emission time	~ 1.187 Gly post-Big-Bang
Comoving radial distance	~ 82–84.8 Gly
Light-travel time	~ 24 Gly
Space expansion factor	× 12.1 to ~ 17.73 Gly

JWST boundary prediction. The TauTimp framework predicts that **JWST cannot detect any object at comoving distance exceeding $R_{LU} = 17.72738$ Gly** in the rest frame of light emission. Galaxies at higher redshift — appearing further only due to space expansion — populate the observable universe out to ~ 93 Gly, but none originate from beyond the Local Universe. This is a **falsifiable prediction**: any detection of a causally connected object beyond $2R_s$ in the rest frame would contradict the muon onduscular horizon postulate.

6 Part V — Wheeler's Delayed-Choice Experiment and the Transactional Interpretation of Quantum Mechanics

Primary sources cited in the original note: A. G. Manning, R. I. Khakimov, R. G. Dall, A. G. Truscott — *Wheeler's delayed-choice gedanken experiment with a single atom*, Nature Physics (local reference nphys3343.pdf); J. G. Cramer — *The Transactional Interpretation of Quantum Mechanics and Quantum Nonlocality*, arXiv:1503.00039.

Wave and particle behaviour are two mutually exclusive-seeming manifestations of light and matter, and the fact that the choice of measurement determines which of the two is observed remains one of the most counter-intuitive features of quantum mechanics — the same duality treated conceptually in Part III. This part revisits Wheeler's delayed-choice gedanken experiment — realised with a single ultra-cold metastable helium atom in a Mach–Zehnder interferometer by Truscott and collaborators — through the lens of John Cramer's Transactional Interpretation of Quantum Mechanics (TIQM), in which a quantum process is mediated by a retarded *offer wave* and an advanced *confirmation wave* forming a single atemporal *transaction*. Within the TauTimp framework, two explicit, distinct propagation speeds are assigned to the two wave sectors of the model: the **offer wave** (the physical, detectable photon or atom) propagates at the ordinary speed of light c, while the **confirmation wave** — identified here with the longitudinal *entanglement particle* that mediates nonlocal correlations between entangled quanta — propagates at $V = c \times 10^9$, the same superluminal interaction speed already adopted for the TauTimp gamma field in TauTimp_Gamma_Wave2.md. The distinction is essential: it keeps every directly observable signal strictly luminal, while confining the superluminal sector to a channel that carries no independently readable message.

6.1 19. Wave–Particle Duality and the Role of Measurement

Wave and particle behaviour are both natural manifestations of light and matter; the choice of measurement apparatus determines which of these seemingly incompatible behaviours is observed — a defining example of the counter-intuitive character of quantum mechanics.

Double-slit experiment. Photons interfere because they are temporarily ionised by tau-neutrino particles. The temporarily ionised photons can, in this picture, pass through the slits one at a time as a wave, and still build up the interference pattern.

Mach–Zehnder interferometer. Temporarily ionised photons travel, one at a time, along two paths — 50% one way, 50% the other. If the temporal separation between successive photons is shorter than one time quantum, they interfere with each other.

6.2 20. Temporal Ionisation and Neutrino Energy Thresholds

Nearly all particles carry spin (spin angular momentum — $\hbar/2$ for the electron) and are entangled through a particle-mediated **longitudinal wave** that propagates faster than light. In this formulation the propagation speed is fixed at

$$\boxed{V = c \times 10^9 = 2.998 \times 10^{17} \text{ m/s}} \quad (31),$$

and $V = c \times 10^9$ is retained here as the single, consistent value used throughout the TauTimp corpus. The wavefunction collapses at measurement with photons in the optical window 400–700 nm, energy 2–4 eV — an energy that blocks the wave-to-particle transition because it exceeds the tau-neutrino threshold value of 0.925 eV.

Measuring a single particle with an infrared sensor, at photon energy $h\nu$ below 0.925 eV, does not produce complete temporal ionisation, and generates no muon-neutrino unless the energy exceeds the threshold by at least 0.018 eV.

6.3 21. Wheeler's Delayed-Choice Experiment

The phenomenon above is illustrated by Wheeler's famous delayed-choice gedanken experiment. It was recently realised, in a single-photon-equivalent configuration, using a single ultra-cold metastable helium atom in a Mach–Zehnder interferometer — an atomic analogue of Wheeler's original proposal (Truscott, Manning, Khakimov & Dall).

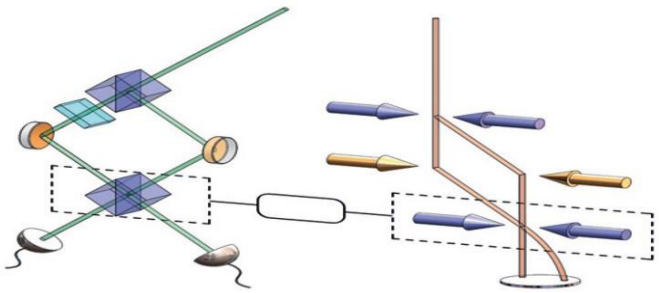


Fig. 1 — Schematic of Wheeler's delayed-choice experiment: the optical Mach–Zehnder interferometer (left) and its atomic analogue (right), after Manning, Khakimov, Dall & Truscott, Nature Physics (local reference nphys3343.pdf).
Truscott's experiment (2015). A single helium atom is sent into an interferometer, and the decision to insert or remove the second beamsplitter — i.e. whether interference is permitted — is made *after* the atom has already entered the interferometer.

6.4 22. The Transactional Interpretation of Quantum Mechanics (TIQM)

The essence of the Transactional Interpretation, as formulated by Cramer, is that a quantum process (e.g. the emission and absorption of a photon) occurs through an exchange of waves between an emitter and a receiver:

- The **retarded wave** (the ordinary wave) is sent forward in time by the source.
- The **advanced wave** is sent backward in time by the absorber.
- The two waves jointly form a **transaction**, a timeless agreement between cause and effect.

Following Cramer's original construction, with one explicit adjustment adopted here: the advanced wave, sent backward in time, is a **longitudinal wave** — the entanglement particle — and therefore propagates at $V = c \times 10^9$, not at an unspecified "much higher than c" rate.

Fig. 2a — Space-time diagram of the transaction: the retarded offer wave (source to transaction) and the advanced confirmation wave (transaction to detector).

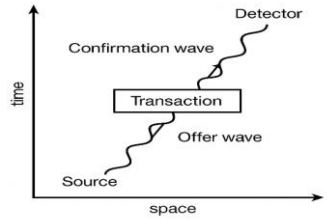
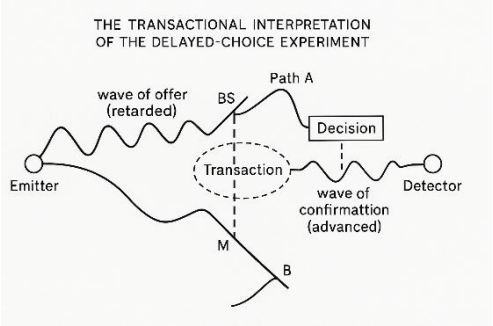


Fig. 2b — The Transactional Interpretation applied to the delayed-choice experiment: the offer wave (retarded) reaches the beamsplitter (BS); the confirmation wave (advanced) returns from the detector after the decision is registered, closing the transaction between emitter and detector via the mirror (M) / beamsplitter (B) paths.



Born rule: $P = |\psi(r,t)|^2 = \psi(r,t) \cdot \psi^*(r,t)$ ⇒ Transaction $\sim \psi \cdot \psi^*$

Transactional framework.

- **Offer wave** ψ_O: propagates forward in time, from the emitter to all possible detectors, at speed c.
- **Confirmation wave** ψ_C: propagates backward in time, from a detector that might register the particle — sent only if absorption is possible — at speed $V = c \times 10^9$.
- **Transaction**: a standing wave formed between the offer and the confirmation wave, determining where the particle actually manifests.

Superluminal advanced-wave assumption. The confirmation wave travels, at $V = c \times 10^9$, effectively instantaneously back to the emitter on laboratory timescales. This eliminates any measurable time lag in the emitter's "knowledge" of the future boundary conditions set at the detector.

6.5 23. Mathematical Model

Offer wave from the emitter. The atom's wavefunction splits into two interferometer paths:

$$\psi_O = \frac{1}{\sqrt{2}}[\psi_A(x,t) + \psi_B(x,t)]$$

Each component acquires a phase set by the potential $V_A(x)$, resp. $V_B(x)$, along its path:

$$\psi_A(x,t) = e^{i(\varphi_A - \omega t)}, \qquad \varphi_A = \int_0^L k_A(x)\,dx, \qquad k_A(x) = \frac{\sqrt{2m(E - V_A(x))}}{\hbar}$$

$$\psi_B(x,t) = e^{i(\varphi_B - \omega t)}, \qquad \varphi_B = \int_0^L k_B(x)\,dx$$

Confirmation wave from the detector. Depending on whether the second beamsplitter is inserted, the detector sends back a **coherent superposition** (enabling interference) or a **single-path response** (disabling interference):

$$\psi_C = \frac{1}{\sqrt{2}}\left[e^{-i\varphi_A} + e^{-i\varphi_B}\right] \qquad \text{(interference enabled)}$$

$$\psi_C = e^{-i\varphi_A} \qquad \text{(interference disabled)}$$

This wave propagates back at $V = c \times 10^9$, letting the emitter effectively "know" the experimental configuration on a timescale far shorter than the light-crossing time of the apparatus.

The transaction:

$$T = \psi_O \cdot \psi_C = \begin{cases} -\frac{1}{2}\left[1 + e^{i(\varphi_A - \varphi_B)}\right] & \text{(interference possible)} \\ -\frac{1}{\sqrt{2}} & \text{(no interference)} \end{cases}$$

Interference condition. The phase difference $\Delta\varphi = \varphi_A - \varphi_B$ determines constructive or destructive interference. **No interference:** the transaction amplitude is then independent of the relative phase; the two paths are fully distinguishable.

6.6 24. Physical Consequences

- The emitter receives effectively instantaneous feedback, via the advanced confirmation wave, regarding whether interference is permitted.
- Even though the choice is made after emission, the transaction forms only if there is consistent phase matching between the offer and confirmation waves.
- Retrocausality in the ordinary sense is avoided by treating the entire process as an atemporal negotiation among all boundary conditions, rather than a sequence of causally ordered events.

6.7 25. Key Insight

The particle's "decision" to behave as a wave or as a particle is not made *at* the beamsplitter or the detector, but nonlocally and atemporally, through the matching of the offer and confirmation waves across the entire spacetime configuration of the experiment — including settings fixed only in the future.

Superluminal transactional laws. All particles communicate bidirectionally in time through an otherwise undetectable channel, at $V = c \times 10^9$ for the confirmation/entanglement sector — never for the directly observable photon, which remains bound to c.

Atemporal partial determinism. Past and future are co-determined; events "settle" according to overall coherence across the full boundary-value problem, not by strict forward propagation alone.

Active non-locality. Distant events can influence a local transaction through this invisible channel.

Locally reversible causality. Effects may appear to precede their apparent cause, but only insofar as this is consistent within a single, completed transaction.

6.8 26. Summary of Numerical Values

Quantity	Symbol	Value
Speed of light (offer/retarded wave, the observable photon)	c	2.998×10^8 m/s
Entanglement-particle speed (confirmation/advanced wave)	V	$c \times 10^9 = 2.998 \times 10^{17}$ m/s
Tau-neutrino energy threshold	—	0.925 eV
Additional threshold for muon-neutrino generation	—	+0.018 eV
Optical collapse window	—	400–700 nm (2–4 eV)

This regenerated version fixes that value at the single TauTimp-canonical figure $V = c \times 10^9 \approx 2.998 \times 10^{17}$ m/s (already adopted in TauTimp_Gamma_Wave2.md for the gamma field), so that every document in this corpus refers to the same numerical constant. The speed of light c is left untouched throughout for every directly observable wave (the offer wave, i.e. the physical photon or atom); only the advanced/confirmation-wave sector — which carries no independently readable message, per the no-signalling requirement discussed in Rezumat_Pettini_Entanglement_RO.md — is assigned the superluminal value.

7 Part VI — Parameter Tables

7.1 27. Fundamental TauTimp Cosmological and Field Parameters

Symbol	Value	Units	Description
R	4.2355×1022	m	TauTimp graviton/electron/photon radius ($\approx 4.48\ Mly$)
R_s	8.3855194×1025	m (= 8.863249 Gly)	Muon onduscular radius

Symbol	Value	Units	Description
$2R_s$	1.6771039×1026	m (= 17.72738 Gly)	Local Universe boundary
R_s/R	1979.8	—	Gravity-to-space scaling ratio
$r_{\min}$	2.7437×10^{24}	m (= 0.290 Gly)	Inner boundary of the flux shell
c	2.99792458×10^{8}	m/s	Speed of light
H_0	67.41 (Planck 2018)	km/s/Mpc	Hubble constant
c/H_0	1.37229×10^{26}	m (= 14.505 Gly)	Hubble radius R_H

7.2 28. Primordial Black Hole Parameters

Symbol	Value	Units	Description
$R_{\text{Sch}}^{(\text{PBH})}$	$\approx R$	m	PBH Schwarzschild radius ($\approx 4.48\ Mly$)
M_{PBH}	$\approx 1.31 \times 10^{19}$	$M_\odot$	PBH mass
$r_{\text{PBH}}^{(\text{linear})}$	~ 1663	m ($\sim$ 1 mile)	PBH linear dimension
ρ_{PBH}	$\sim M_{\text{PBH}}/r_{\text{PBH}}^3$	kg/m^3	PBH density (uniform quark soup)
$t_\star$	290–312	Myr	First-star epoch
t_U	13.79	Gyr	Local Universe age

7.3 29. Galactic Black Hole (Sagittarius A*) Parameters

Symbol	Value	Units	Description
$M_{\text{Sgr A*}}$	4×10^6	$M_\odot$	Mass of Sgr A*
$r_s^{\text{Sgr A*}}$	1.217×10^{10}	m	Sgr A* Schwarzschild radius
$R_{\tau\text{EvHo}}$	75.587	m	Temporal Event Horizon radius
$R_{s\text{EvHo}}$	1663	m	Space-collapse / inflation trigger radius
$\ell_{\min}^{\text{Sgr A*}}$	3.524×10^{-25}	m	Minimum inter-quark distance
$m_q^{\text{Sgr A*}}$	2.5559×10^{-31}	kg	Per-quark mass at centre
τ_{cross}	1.17548×10^{-33}	s	Light-crossing time of $\ell_{\min}$

7.4 30. Quantum Limits and Wheeler/TIQM Values

Symbol	Value	Units	Description
ℓ_q	0.43×10^{-18}	m	Quark linear dimension (spatial quantum)
d_N	0.841419×10^{-15}	m	Nucleon (proton) charge radius
d_N/d_q	1957.0	—	Nucleon-to-quark size ratio
T_q	$\sim 1.765 \times 10^{-19}$	s	Photon temporal quantum
a_0	5.29177×10^{-11}	m	Bohr radius
Experimental time resolution	$\sim 10^{-18}$	s	Current best
Planck time t_P	5.39×10^{-44}	s	For comparison
V (entanglement/confirmation-wave speed)	$c \times 10^9 = 2.998 \times 10^{17}$	m/s	Superluminal, no-signalling sector
Tau-neutrino energy threshold	0.925	eV	Collapse condition
Muon-neutrino generation threshold (additional)	+0.018	eV	On top of the tau-neutrino threshold
Optical collapse window	400–700 (2–4 eV)	nm	Measurement window

7.5 References

1. Planck Collaboration (2018). *Planck 2018 results. VI. Cosmological parameters.* A&A 641, A6.
2. Oesch, P. A. et al. (2016). *A remarkably luminous galaxy at z = 11.1.* ApJ 819, 129. [GN-z11]
3. Wright, E. L. (2006). *A Cosmology Calculator for the World Wide Web.* PASP 118, 1711. https://www.astro.ucla.edu/~wright/CosmoCalc.html
4. arXiv:1604.01280 — Quark size and substructure limits.
5. Wikipedia. *GN-z11.* https://en.wikipedia.org/wiki/GN-z11
6. Hartle, J. B. & Hawking, S. W. (1983). *Wave function of the universe.* Phys. Rev. D 28, 2960.
7. Bannikov, V. V. (2017). Private communication / ResearchGate — On the experimental detectability of the temporal quantum.
8. Manning, A. G., Khakimov, R. I., Dall, R. G., Truscott, A. G. — *Wheeler's delayed-choice gedanken experiment with a single atom*, Nature Physics (local reference nphys3343.pdf).
9. Cramer, J. G. — *The Transactional Interpretation of Quantum Mechanics and Quantum Nonlocality*, arXiv:1503.00039.
10. Popescu, V.-M. — *TauTimp — Gamma-Ray Longitudinal Wave: Physical Foundations and Full Field-Theoretic Derivation*, TauTimp_Gamma_Wave2.md (source of $V = c \times 10^9$).

11. Popescu, V.-M. (2025–2026). *TauTimp Unified Framework — Derivation of the Spherical Onduscular Wave Equation, GEO/Mars Temporal Flux Constraints, and Muon Neutrino Postdiction of H_0.* Unpublished manuscript.
12. Popescu, V.-M. (2025–2026). *TauTimp Cosmology of the Local Universe and the Measurement Problem.* Unpublished manuscript.

Document: TauTimp / Teoria Onduscularǎ — Quantization of Space, the Primordial Black Hole, Muon Neutrino Flux, and Wheeler's Delayed-Choice Experiment Author: Viorel-Mihai Popescu | Version: v2.0 — 21 August 2026 Compatible with TauTimp Unified Framework v1.0 and TauTimp Cosmology of the Local Universe v1.0